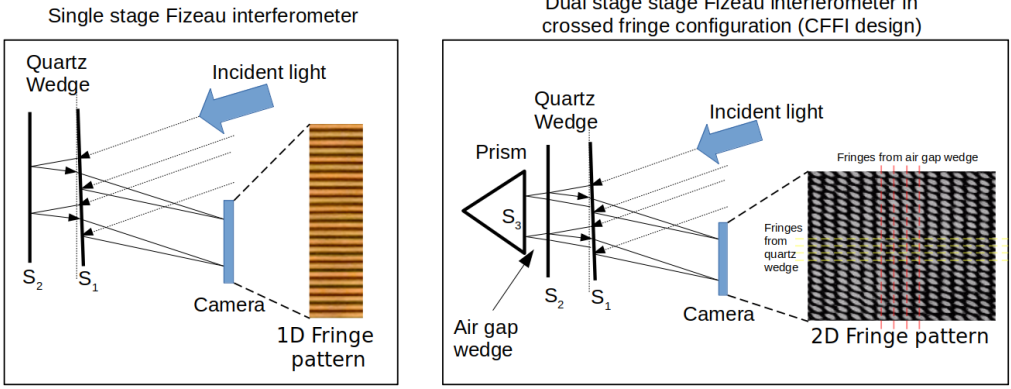

Figure 12: (Left) A schematic diagram illustrating the working principle of a single-stage Fizeau interferometer. The partial reflections of the incident light on the front and back surface (S1 and S2) of a quartz wedge interfere and give rise to an interference pattern consisting of a series of evenly spaced bright and dark fringes (1D fringe pattern). (Right) Working principle of the CFFI design with a 2-stage Fizeau interferometer. Here an additional surface S3 (face of an equilateral prism) is placed behind the quartz wedge creating a second wedge (air gap wedge). The quartz and the air gap wedge are oriented orthogonal to each other and the resulting interference pattern resembles a grid of evenly spaced bright spots (2D fringe pattern). This 2D pattern is a product of the two orthogonal 1D patterns generated from the quartz (parallel to dashed yellow lines) and air gap wedge (parallel to the dashed red lines). The wavelength of the incident light is estimated by analyzing the 2D interferogram captured by a camera